

## Unit 3, Lesson 8: Angle Relationship Proofs – Part 1



### Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



### Learning Target

- ☐ I can prove relationships using parallel lines and their transversals.



### 3.8.1 Warm-Up: Would You Rather?

- The meals at two restaurants range from \$8.00 to \$25.00. Servers are paid based on the following.

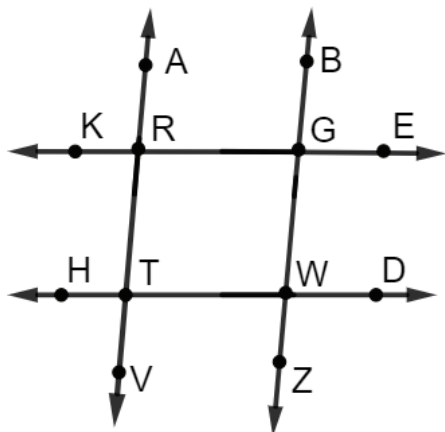
| <u>Restaurant A</u> | <u>Restaurant B</u> |
|---------------------|---------------------|
| \$18 per hour       | \$10.50 per hour    |
| No tipping allowed  | Tipping encouraged  |

Would you rather work as a server at Restaurant A or at Restaurant B? Justify your choice.



### 3.8.2 Collaboration: Parallel Lines

1. Contradictory information is given for the figure.  
 $\overleftrightarrow{KE} \parallel \overleftrightarrow{HD}$ ,  $\angle KRA \cong \angle BGE$ ,  $\angle TRG$  is a supplement of  $\angle RGW$ ,  
 and  $m\angle DWZ = 87^\circ$ .



- What piece of information is contradictory?
- How do you know it is contradictory?
- Share your contradiction with your partner. Discuss any differences.

- d. Create four different pieces of information about this figure using  $\overleftrightarrow{AV} \parallel \overleftrightarrow{BZ}$ . One of the pieces of information should be contradictory.

1.

2.

3.

4.

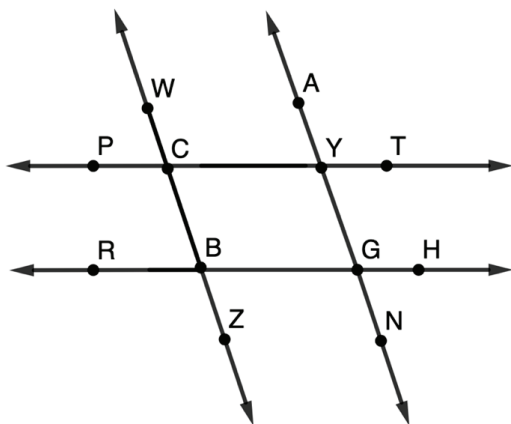
- e. Trade information with a partner and see if your partner can find the contradiction. Write their contradiction in the table.

| Partner's Contradiction | Partner's Initials |
|-------------------------|--------------------|
|                         |                    |

2. Terrence was organizing a card sort for the proof, but his cards fell on the floor and are now out of order.

Given:  $\overrightarrow{PT} \parallel \overrightarrow{RH}$  and  
 $\angle CBG \cong \angle CYG$

Prove:  $\overrightarrow{WZ} \parallel \overrightarrow{AN}$



| Statement                                              | Reason                                                                                                                               |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| A. $m\angle CBG = m\angle CYG$                         | K. Definition of Supplementary Angles                                                                                                |
| B. $\angle CBG$ and $\angle BGY$ are supplementary     | L. Substitution Property of Equality                                                                                                 |
| C. $\angle CBG \cong \angle CYG$                       | M. If two lines are intersected by a transversal so that consecutive interior angles are supplementary, then the lines are parallel. |
| D. $\angle CYG$ and $\angle BGY$ are supplementary     | N. Definition of Congruence                                                                                                          |
| E. $\overrightarrow{WZ} \parallel \overrightarrow{AN}$ | P. Given                                                                                                                             |
| F. $m\angle CBG + m\angle BGY = 180^\circ$             | R. If two parallel lines are intersected by a transversal, then the consecutive interior angles are supplementary.                   |
| G. $\overrightarrow{PT} \parallel \overrightarrow{RH}$ | S. Definition of Supplementary Angles                                                                                                |
| H. $m\angle CYG + m\angle BGY = 180^\circ$             | T. Given                                                                                                                             |

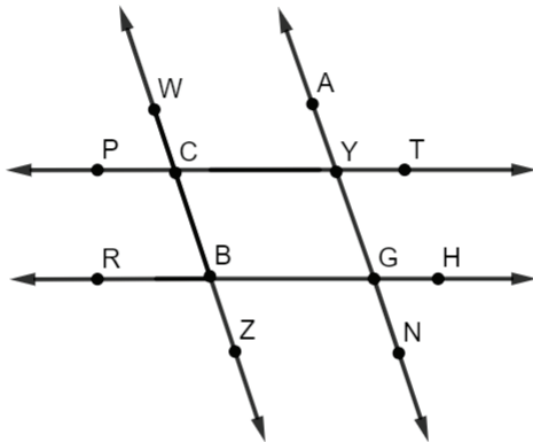
Work with your partner to correctly place the statements and reasons in the correct order by writing the letter.

| Statement | Reason |
|-----------|--------|
| 1.        | 1.     |
| 2.        | 2.     |
| 3.        | 3.     |
| 4.        | 4.     |
| 5.        | 5.     |
| 6.        | 6.     |
| 7.        | 7.     |
| 8.        | 8.     |

3. Work with your partner to complete the proof in a different way.

Given:  $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$  and  $\angle CBG \cong \angle CYG$

Prove:  $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$



| Statement                                                      | Reason                                                                                              |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1. $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$ | 1. Given                                                                                            |
| 2. $\angle CBG \cong \angle \underline{\hspace{2cm}}$          | 2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent. |
| 3. $\angle CBG \cong \angle CYG$                               | 3. Given                                                                                            |
| 4.                                                             | 4. Transitive Property of Congruence                                                                |
| 5. $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$ | 5.                                                                                                  |

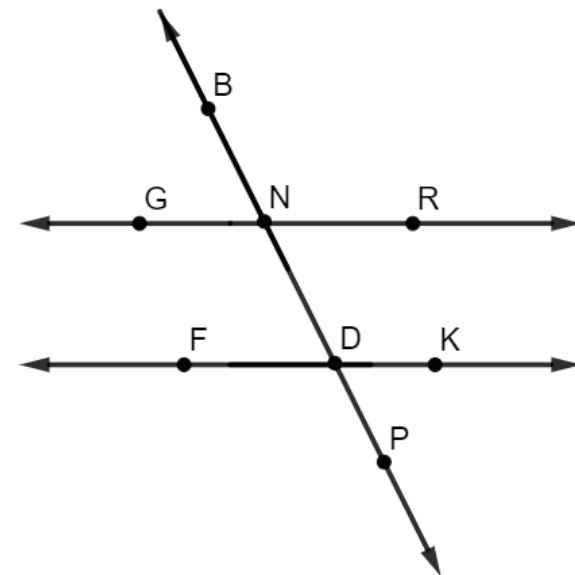


### 3.8.3 Guided Instruction: Proofs Involving Angle Relationship

1. Complete the flowchart proof.

Given:  $\angle RND$  and  $\angle FDP$  are supplementary.

Prove:  $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$



$\angle RND$  and  $\angle FDP$  are supplementary.  $\rightarrow$

$\angle FDP \cong \angle NDK$   $\rightarrow$

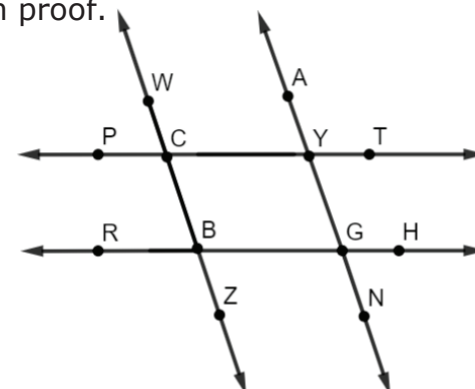
$\rightarrow$   $\angle RND$  and  $\angle NDK$  are supplementary.

Substitution

2. Choose from the bank of statements and reasons to complete the paragraph proof.

Given:  $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$  and  $\angle PCW \cong \angle HGN$

Prove:  $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$



It is given that \_\_\_\_\_,

Therefore  $\angle RBC \cong \angle PCW$  because if two parallel lines are intersected by a transversal then \_\_\_\_\_.

It is also given that \_\_\_\_\_. By the \_\_\_\_\_,

it can be written that  $\angle RBC \cong \angle HGN$ . Therefore, it can be concluded that  $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$  because if two lines are intersected by a transversal so that \_\_\_\_\_ then the lines are parallel.

#### Statements and Reasons

- |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• <math>\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}</math></li> <li>• <math>\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}</math></li> <li>• <math>\angle PCW \cong \angle HGN</math></li> <li>• <math>\angle ZBG \cong \angle HGN</math></li> <li>• <math>\angle GND</math></li> <li>• <math>\angle FDP</math></li> </ul> | <ul style="list-style-type: none"> <li>• Definition of Congruence</li> <li>• Transitive Property of Congruence</li> <li>• Definition of Supplementary Angles</li> <li>• Corresponding angles are congruent.</li> <li>• Alternate exterior angles are congruent.</li> <li>• Alternate interior angles are congruent.</li> </ul> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



### 3.8.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help someone.

|                                                                            |                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Using <i>substitution</i> as a reason in a proof.                          |    |
| Using the <i>transitive property of congruence</i> as a reason in a proof. |    |
| Using the <i>definition of congruence</i> as a reason in a proof.          |    |
| Understanding what angles are congruent when two lines are parallel.       |    |
| Understanding what angles are supplementary when two lines are parallel.   |    |



## Unit 3, Lesson 9: Angle Relationship Proofs – Part 2



### Benchmarks

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